

AHMED ELMAHDY

Platform/DevOps Engineer | Azure | AWS | Terraform | Kubernetes | CI/CD

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Professional Summary

Platform & DevOps Engineer with 5+ years of experience delivering cloud infrastructure, CI/CD, Kubernetes platforms, and environment modernization across Azure and AWS, Strong background in inherited-platform recovery, Terraform-based infrastructure management, microservice onboarding, secure delivery patterns, and cloud cost optimization, Proven ability to reduce deployment risk (0 staging IaC drift), improve platform reliability (24 HPA policies across 4 environments), and drive operational clarity through documentation and stakeholder communication.

Core Skills

- **Cloud & Infrastructure:** Azure, AWS, Linux Admin, Cloud Environment Assessment, Platform Recovery
- **Infrastructure as Code:** Terraform, CloudFormation, Drift Analysis, Environment Alignment, TF State Management
- **Containers & Orchestration:** Docker, Kubernetes, Helm, Argo CD, AKS, Multi-Environment Deployments
- **CI/CD & Automation:** GitHub Actions, Jenkins, GitLab-to-GitHub Actions Migration, Bash, Python.
- **Identity & Security:** Azure Entra ID, RBAC, Key Vault, IAM roles, group-based access control, secret governance.
- **Observability & Operations:** Prometheus, Grafana, CloudWatch, Azure monitoring/logging, debugging, runbooks.
- **Cost & Governance:** Cost Management, right-sizing, tagging/governance, orphaned-resource, infrastructure assessment.
- **Documentation & Delivery:** Technical Reporting, Architecture Diagrams, Confluence, Remediation Planning, Stakeholder Communication

Professional Experience

➤ Platform/DevOps Engineer | Julfar Consultancy Services | Jan 2026 – Present

- Stabilized inherited Azure/Terraform platforms across DEV, STAGING, and PROD by resolving drift, state, and configuration risks and establishing safer infrastructure baselines.
- Onboarded microservice to the ActionAI platform end-to-end: GitHub repos, GitHub Actions CI/CD, Helm charts, and Argo CD GitOps automation across 3 environments.
- Integrated secure delivery patterns across Azure Key Vault, Terraform, APIM, and Azure/Entra RBAC, strengthening secrets governance, service exposure control, and access management.
- Modernized CI/CD by migrating pipelines from GitLab to GitHub Actions and validating reliable execution of build, deployment, and secret-dependent workflows.
- Implemented 24 HPA policies (6 services × 4 environments) with tiered scaling guardrails, optimized workload sizing to 1 vCPU / 2Gi for heavy processing services and achieved 0 staging IaC drift through Terraform alignment.
- Improved maintainability and handover readiness through READMEs, Confluence documentation, architecture diagrams, and stakeholder-facing technical reporting.

➤ **System Administrator | Julfar Consultancy Services | Oct 2021 – Dec 2025**

- Administered on-premises and cloud-based systems, maintaining 99.9% uptime through proactive monitoring and incident response.
- Supported DevOps and development teams by configuring and maintaining CI/CD pipeline environments.
- Automated system maintenance, backups, and deployments using Bash and Python scripts.
- Assisted in AWS service management (EC2, IAM, VPC) and optimized resource utilization.
- Enhanced deployment reliability by reducing manual configuration errors by 20%.
- Collaborated with cross-functional teams to improve system scalability and uptime.

➤ **DevOps Engineer Intern | DEPI – MCIT | May 2024 – Feb 2025**

- Gained hands-on experience with AWS, Docker, Kubernetes, Terraform, Ansible, and Jenkins.
- Built and optimized CI/CD pipelines for application deployment and infrastructure automation.
- Managed infrastructure provisioning using Terraform and CloudFormation (IaC).
- Monitored application performance with Prometheus and Grafana integrated with CloudWatch.
- Worked in Agile teams to implement real-world DevOps practices and continuous improvement.

See implementation details in portfolio: <https://ahmed-el-mahdy.github.io/#architecture>

➤ **AWS Cloud DevOps Nanodegree (MCIT & Udacity) | Nov 2022 – Apr 2023**

- Designed and deployed AWS infrastructure using Terraform and CloudFormation.
- Built CI/CD pipelines for automated deployment to EC2 and S3.
- Integrated observability tools to ensure system reliability and performance.
- Applied Agile principles for team collaboration and project delivery.

See implementation details in portfolio: <https://ahmed-el-mahdy.github.io/#architecture>

➤ **Infra & Systems Engineer | Shatta Company| Feb 2019 – Mar 2020**

- Administered multi-OS environments, Proxmox clusters, VLANs, and firewalls to ensure system security.
- Implemented and managed VPN connections for secure multi-site connectivity.
- Enhanced infrastructure efficiency through proactive monitoring and maintenance.

Certifications

- Microsoft Identity and Access Administrator Associate (SC-300) 2025
- **AWS Certified Solutions Architect Associate (2025)**
- **AWS Certified Cloud Practitioner (2025)**

Professional development and training

- Azure Administrator Associate Training (2025)
- **CKA Certification Course** — KodeKloud (2025)
- Advanced Cloud DevOps **Nanodegree** from Udacity & MCIT (2023)
- **McKinsey Forward Program** (2025)

PROJECTS

- **Mosadad Recovery Claim Platform - Azure, Terraform, GitHub Actions, Kubernetes (2026)**
Operated 4 AKS clusters running 7 .NET 8 microservices and 2 Angular 19 frontends across 4 environments (DEV/QA/STAGING/PROD), Implemented 24 HPA policies with tiered scaling and tuned Claims service to 1 vCPU / 2Gi for PDF processing stability, Achieved 0 staging IaC drift through full Terraform alignment, Built DRY GitHub Actions CI/CD: Docker → ACR → AKS → APIM auto-import. Configured production monitoring with 4 alert rules + 3 availability tests via Grafana, Prometheus, and Application Insights.
- **ActionAI Platform Enablement-Azure, GitHub Actions, Helm, Argo CD, APIM, Key Vault (2026)**
Delivered end-to-end onboarding of new microservices through repository setup, GitHub Actions CI/CD, Helm-based multi-environment deployment design, and Argo CD post-sync integration. Automated service exposure and platform integration through Azure API Management endpoint registration and environment-specific cluster deployment patterns. Strengthened secure delivery with Azure Key Vault integration, Terraform-based secret governance, and Azure/Entra group-based RBAC.
- **Serverless Image Processing - AWS S3, Lambda (2025)**
Built a serverless image-processing pipeline using S3 and Lambda to reduce operational overhead-40%.
- **N8N + Nginx + Ngrok Automation (2025)**
Built containerized workflow automation with reverse proxy, SSL, and automated ngrok URL synchronization for webhook continuity.
- **DevOps CI/CD Project (2024)**
Automated deployment of a 3-tier app on AWS using Jenkins, Terraform, and EKS; achieved auto-scaling and high availability.
- **Shell-Scripted Database Management System – DEPI & MCIT (2024)**
Developed a CLI-based DB management system with shell script for local data storage and query operations.
- **Udapeople CI/CD App – MCIT & Udacity (2023)**
Deployed a dynamic website using CloudFormation and CircleCI integrated with AWS services.
- **Infrastructure Automation – MCIT & Udacity (2022)**
Automated AWS static site infrastructure using CloudFormation templates to ensure repeatable provisioning.

Education

- **B.Sc. in Computer Science, Delta University, GPA 3.7 (2018)**
- **ITI Diploma – System & Network Administration (2019)**

Languages:

Arabic (Native Language) English (Professional Working Proficiency)